

C²-SFL: Class-Balanced and Cost-Aware Split Federated Learning for Mobile Edge Computing

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Abstract

Split Federated Learning (SFL) facilitates collaborative training in Mobile Edge Computing (MEC) by splitting models between clients and servers. However, existing client selection and resource allocation strategies fail to account for data and system heterogeneity, thereby exacerbating data skew and leading to inefficient training and high costs. To address these issues, we propose a Class-Balanced and Cost-Aware SFL (C²-SFL) framework that jointly optimizes client selection, model partitioning, and bandwidth allocation. We introduce a Relation-Aware Class-Balanced Algorithm (RACBA) to select clients by balancing data distribution and energy states, along with a Cost-Driven Optimization Algorithm (CDOA) to dynamically adjust model split points and bandwidth. Experiments show that C²-SFL significantly reduces training costs, accelerates convergence, and improves accuracy under heterogeneous MEC settings, achieving reductions of 12.6%-71.5% in energy consumption and 5.2%-50.9% in training latency.

CCS Concepts

• Computer systems organization → Client-server architectures.

Keywords

Split Federated Learning, Data Heterogeneity, Client Selection, Model Partitioning, Bandwidth Allocation

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1 Introduction

The exponential growth of data from mobile clients in Mobile Edge Computing (MEC) highlights the limitations of centralized Machine Learning, including its high communication overhead, resource consumption, and data privacy risks[1, 2]. Split Federated Learning (SFL)[3] mitigates these limitations by combining Split Learning (SL) and Federated Learning (FL) through model partitioning, which allows collaborative training between edge devices and servers. This framework reduces computational burdens on clients and preserves data privacy, making it suitable for MEC environment[4].

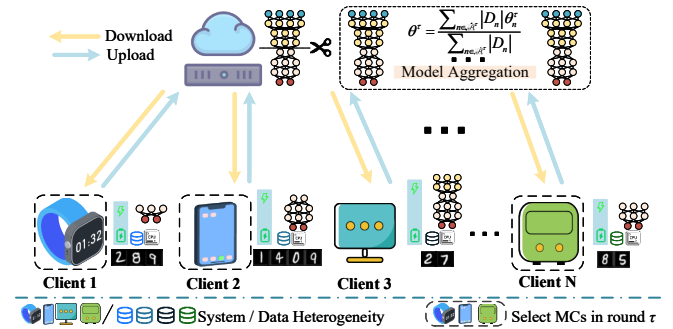


Figure 1: SFL model in a MEC Network. The MEC network consists of M MCs and a PS, where the MCs have system heterogeneity and data heterogeneity.

However, the application of SFL in heterogeneous MEC networks faces two major challenges. **1) First**, non-IID data distributions induce client drift and slow convergence, a problem exacerbated by uncoordinated client selection, which thus compromises convergence efficiency and stability[5, 6]. **2) Second**, system heterogeneity directly constrains the practical efficiency and scalability of SFL in dynamic environments, as it complicates the joint optimization of client selection, model splitting, and resource allocation[7].

Motivated by the above challenges and the impact of data heterogeneity on convergence, we propose C²-SFL (Class-Balanced and Cost-Aware Split Federated Learning), a unified framework that co-designs client selection, model partitioning, and bandwidth allocation for efficient SFL in heterogeneous MEC networks.

The main contributions are summarized as follows:

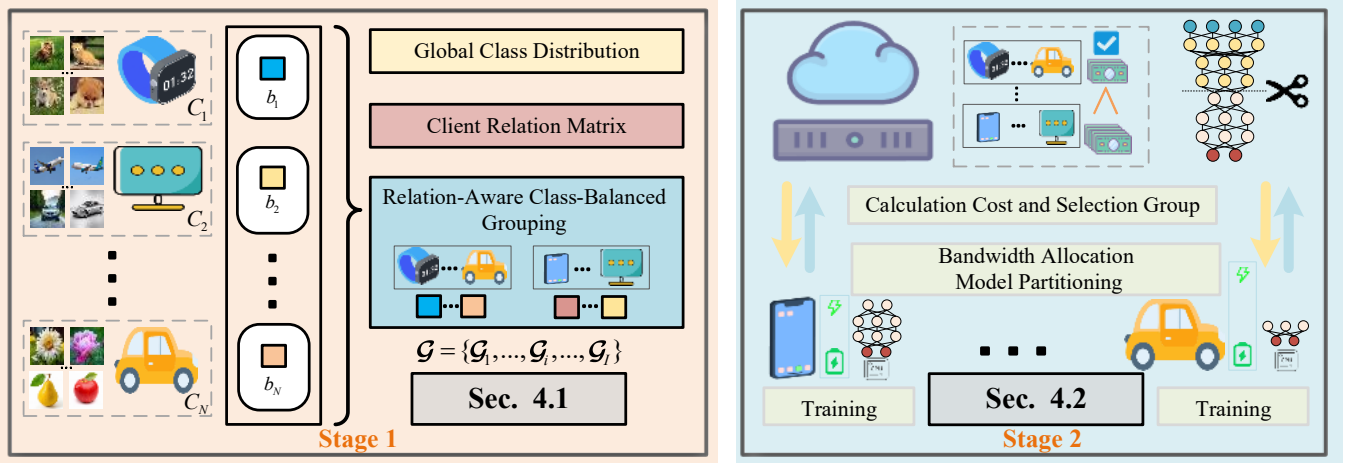


Figure 2: The overview of Class-Balanced and Cost-Aware SFL (C²SFL). Our C²-SFL framework operates in two stages: (1) Relation-aware class-balanced client grouping based on data distributions; (2) Joint optimization of model partitioning and bandwidth allocation to minimize each latency and energy consumption.

- We propose C²-SFL, a novel framework that jointly optimizes client selection, model partitioning, and bandwidth allocation to minimize training cost in heterogeneous MEC networks. This is achieved by decomposing the problem into a holistic subproblem of global convergence rounds and a partial subproblem of costs per round.
- We introduce a Relation-Aware Class-Balanced Algorithm (RACBA) to adaptively select clients with complementary data distributions, mitigating the impact of non-IID data. Additionally, we develop a Cost-Driven Optimization Algorithm (CDOA) to dynamically configure model partitioning and bandwidth, thereby enhancing energy efficiency and accelerating convergence.
- Extensive experiments on benchmark datasets, demonstrate that C²-SFL significantly reduces the overall training cost, accelerates convergence speed, and improves model accuracy compared to state-of-the-art algorithms.

2 System Model

As shown in Fig. 1, we combine the considered MEC into one SFL system. We consider an MEC network consisting of one Parameter Server (PS) and N Mobile Clients (MCs). MCs are indexed by $\mathcal{N} = \{1, 2, \dots, N\}$. The MC n has a local dataset $\mathcal{D}_n = \{(x_i, y_i)\}_{i=1}^{|\mathcal{D}_n|}$, $n \in \mathcal{N}$, where x_i and y_i are the features and label.

2.1 Latency Model

For the τ -th round, the set of selected MCs is denoted by $\mathcal{A}^\tau$. The overall latency $T_{\mathcal{A}^\tau}$ is the maximum delay across all MCs. The training delay $t_{n,\tau}^{\text{train}}$ encompasses both the local model segment training on the device and the remaining segment training on the server. Then, $T_{\mathcal{A}^\tau}$ is formulated as follows:

$$T_{\mathcal{A}^\tau} = \max_{n \in \mathcal{A}^\tau} \left\{ \left(t_{n,\tau}^{\text{train}} + t_{n,\tau}^{\text{mid}} \right) \cdot O_n + t_{n,\tau}^{\text{up}} \right\}, \quad (1)$$

where $t_{n,\tau}^{\text{mid}}$ is the intermediate transmission delay, $t_{n,\tau}^{\text{up}}$ is the model upload delay, and O_n is the local iterations.

2.2 Energy Model

The energy consumption $E_{\mathcal{A}^\tau}$ is calculated as the sum for all MCs in $\mathcal{A}^\tau$. The communication energy $e_{n,\tau}^{\text{com}}$ accounts for parameter transmission, while the computational energy $e_{n,\tau}^{\text{train}}$ encompasses both local and server-side model training. Therefore, $E_{\mathcal{A}^\tau}$ can be expressed as:

$$E_{\mathcal{A}^\tau} = \sum_{n \in \mathcal{A}^\tau} (e_{n,\tau}^{\text{com}} + O_n \cdot e_{n,\tau}^{\text{train}}). \quad (2)$$

3 Problem Formulation

The goal of this paper is to achieve cost-effective SFL in MEC, aiming to minimize operational costs and guarantee global model convergence under limited energy consumption and resource constraints. The optimization problem is formulated as follows:

$$P_0 : \min_{\{\mathcal{A}, \mathcal{L}, \mathcal{B}\}} \text{COST} = \sum_{\tau=1}^T [\rho \cdot E_{\mathcal{A}^\tau} + (1 - \rho) \cdot T_{\mathcal{A}^\tau}], \quad (3)$$

where $\mathcal{A}$, $\mathcal{L}$, and $\mathcal{B}$ denote the client selection, model partitioning point, and bandwidth allocation, respectively. ρ is weighting factor between latency and energy consumption.

Since the problem P_0 is NP-hard[7], we assume that the training budget for each mobile client in each round follows a uniform distribution. Consequently, we decompose the optimization problem into a holistic and a partial subproblem, aimed at reducing the number of SFL rounds and the cost per round, respectively.

3.1 Subproblem 1

The number of convergence rounds can be reduced by client selection $\mathcal{A}$. Due to limited bandwidth resources, the PS can only accommodate a subset of M clients simultaneously in each round.

Table 1: Accuracy comparison (%) of different SFL frameworks under non-IID data (NID) settings, NID_α means Dirichlet distribution with ratio α. Best in bold and second with underline.

Methods	Fashion-MNIST			EMNIST			CIFAR-10			CIFAR-100		
	NID _{0.1}	NID _{0.2}	NID _{0.5}	NID _{0.1}	NID _{0.2}	NID _{0.5}	NID _{0.1}	NID _{0.2}	NID _{0.5}	NID _{0.1}	NID _{0.2}	NID _{0.5}
SplitFed[3]	77.44	86.60	88.28	75.16	77.59	82.69	73.99	81.58	85.96	43.81	53.72	59.79
FedProx[8]	78.21	87.92	88.76	76.23	79.56	82.91	73.62	83.62	87.31	45.33	52.30	60.12
AdaptSFL[9]	79.70	84.46	87.25	77.14	78.86	83.22	71.91	81.12	86.24	45.04	52.99	59.67
ESFL[10]	77.81	87.02	88.36	76.21	78.51	82.86	74.26	81.98	85.26	43.66	53.61	60.14
Ours w/o RACBA	78.62	86.89	87.92	75.60	78.12	82.50	73.68	80.65	86.10	44.67	53.86	60.26
Ours w/o CDOA	80.14	87.96	89.21	77.42	79.96	83.32	75.87	82.60	89.26	45.32	54.25	62.10
C ² -SFL (Ours)	80.97	88.66	89.32	77.52	80.07	83.87	76.21	83.65	89.78	46.57	55.08	62.27

Table 2: Compares the convergence rate and final accuracy across different datasets. The R_{acc} columns report the minimum communication rounds required to reach the target test accuracy acc , while the ACC columns present the final stabilized test accuracy after training.

Methods	Fashion-MNIST(NID _{0.2})				EMNIST(NID _{0.2})				CIFAR-10(NID _{0.2})				CIFAR-100(NID _{0.2})			
	$R_{20\%}$	$R_{40\%}$	$R_{80\%}$	ACC	$R_{20\%}$	$R_{40\%}$	$R_{75\%}$	ACC	$R_{20\%}$	$R_{40\%}$	$R_{80\%}$	ACC	$R_{20\%}$	$R_{40\%}$	$R_{50\%}$	ACC
SplitFed[3]	3	4	18	86.60	3	9	81	77.59	3	11	97	81.58	12	44	84	53.72
FedProx[8]	3	4	19	87.92	3	9	76	79.56	3	10	95	83.62	12	45	81	52.30
AdaptSFL[9]	2	6	17	84.46	3	8	52	79.96	2	9	96	81.12	17	53	98	52.99
ESFL[10]	3	5	17	87.02	3	9	84	78.51	3	10	99	81.98	13	51	94	53.61
Ours w/o RACBA	3	5	17	86.89	3	10	88	78.12	3	12	98	80.65	13	55	92	53.86
Ours w/o CDOA	2	3	13	87.96	3	9	50	79.96	2	9	88	82.60	8	31	72	54.25
C ² -SFL (Ours)	2	2	12	88.66	3	8	34	80.07	2	7	84	83.65	8	27	66	55.08

The holistic subproblem P_1 is formulated as follows:

$$P_1 : \min_{\{\mathcal{A}\}} \mathcal{T}. \quad (4)$$

3.2 Subproblem 2

After selecting $\mathcal{A}^\tau$, we consider how to reduce the cost in round τ in SFL. Subsequently, the problem reduces to executing cost-effective learning within this group through fine-grained resource. The partial subproblem P_1 is formulated as follows:

$$P_2 : \min_{\{\mathcal{L}, \mathcal{B}\}} \text{COST} = \sum_{\tau=1}^{\mathcal{T}} [\rho E_{\mathcal{A}^\tau} + (1 - \rho) T_{\mathcal{A}^\tau}]. \quad (5)$$

4 Class-Balanced and Cost-Aware SFL

We leverage class balancing and costs awareness through two complementary modules to accelerate model convergence while jointly optimizing resource allocation to reduce system costs overhead. The overview of our proposed C²-SFL is shown in Fig. 2.

4.1 Relation-Aware Class-Balanced Algorithm

We leverage the client relation matrix $\mathbf{A}$ to quantify distributional affinity. Clients with dissimilar distributions are grouped to form balanced ensembles. The relation matrix $\mathbf{A}$ is given as:

$$\mathbf{A}(k, p) = 1 - J(\mathbf{b}_k \| \mathbf{b}_p), \quad (6)$$

where $\mathbf{A}(k, p) \in [0, 1]$ measures the similarity of class distributions between clients k and p , characterized by their data distributions $\mathbf{b}_k$ and $\mathbf{b}_p$, as quantified by the Jensen-Shannon Divergence J .

After obtaining the client relation matrix $\mathbf{A}(k, p)$, we employ a Relation-Aware Class-Balanced Algorithm to construct class-balanced groups $\mathcal{G}$. The pseudo-code of RACBA is provided in Algorithm 1.

Algorithm 1 Relation-Aware Class-Balanced Algorithm

Input: MCs $\mathcal{M} = \{1, 2, \dots, K\}$, client distribution vectors $\{\mathbf{b}_k\}_{k=1}^K$, relation matrix $\mathbf{A}$, group size M , uniform distribution $\mathcal{H}$.
Output: Groups of MCs $\mathcal{G} = \{\mathcal{G}_1, \mathcal{G}_2, \dots, \mathcal{G}_G\}$.
1: $\mathcal{G} \leftarrow \{\}, k \leftarrow 1, \mathcal{M} \leftarrow \{1, 2, \dots, K\}$;
2: **while** $|\mathcal{M}| \neq 0$ **do**
3: $\mathcal{G}_k \leftarrow [\], \mathbf{B}_{\mathcal{G}_k} \leftarrow \mathbf{0}$;
4: $j \leftarrow$ Randomly choose one index j from $\mathcal{M}$;
5: $\mathcal{M}.\text{remove}(j), \mathcal{G}_k.\text{append}(j)$;
6: **while** $|\mathcal{G}_k| < M$ **and** $|\mathcal{M}| > 0$ **do**
7: $j \leftarrow \arg \min_{j \in \mathcal{M}} \left(\sum_{p \in \mathcal{G}_k} \mathbf{A}(j, p) + (\mathbf{b}_j, \mathcal{H}) \right)$;
8: $\mathcal{G}_k.\text{append}(j), \mathcal{M}.\text{remove}(j)$;
9: **end while**
10: $\mathcal{G}[k] \leftarrow \mathcal{G}_k$;
11: $k \leftarrow k + 1$;
12: **end while**
13: **return** $\mathcal{G}$;

4.2 Cost-Driven Optimization Algorithm

For selected groups, we solve bandwidth allocation via Lagrange multipliers, deriving closed-form solutions under budget constraints.

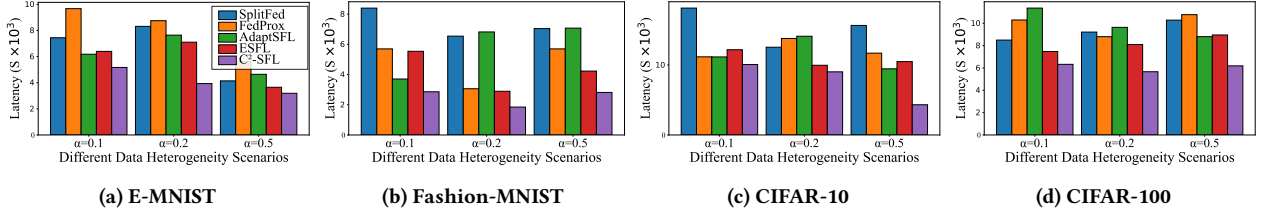


Figure 3: Latency under different data heterogeneity levels and datasets.

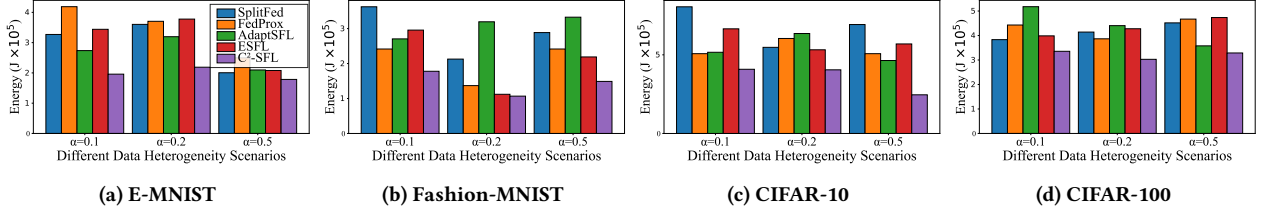


Figure 4: Energy under different data heterogeneity levels and datasets.

Model partitioning is optimized through efficient enumeration of candidate points to minimize training time and energy consumption. This cost-driven alternating optimization algorithm adapts to computational and communication heterogeneity.

5 Performance Evaluation

Configurations: We employ the ResNet-9 model and evaluate it on four distinct datasets. The MEC network architecture consists of one PS and 25 MCs. The PS only can serve up to 5 MCs in each training round.

Baselines: We compare our method with several SOTA SFL methods: standard SFL (SplitFed[3], FedProx[8]), adaptation-based (AdaptSFL[9]), efficiency-based (ESFL[10]).

Accuracy Comparison: As shown in Table 1, we compare the performance of C^2 -SFL with existing SFL baselines under different data heterogeneity settings. Experimental results demonstrate that C^2 -SFL achieves a 2.96% improvement in accuracy on average.

Convergence Rates: As shown in Table 2, C^2 -SFL demonstrates faster convergence compared to baseline methods. With an average reduction of 35.6% in convergence rounds, C^2 -SFL validates the effectiveness of mitigating data heterogeneity.

Cost Comparison: As shown in Figs. 3 and 4, the proposed C^2 -SFL framework consistently outperforms the baselines in both energy consumption and training latency, achieving reductions ranging from 12.6% to 71.5% and from 5.2% to 50.9%, respectively.

Ablation Study: To assess the efficacy of key modules, we perform ablation studies on RACBA and CDOA, with results in Table 1. Convergence rounds directly affect energy consumption and latency overhead and are thus crucial for deployment.

6 Conclusion

This paper proposed C^2 -SFL, an optimization framework for model training in heterogeneous SFL environments. Specifically, to simultaneously mitigate the impact of data heterogeneity and reduce system training costs, our proposed C^2 -SFL outperforms existing

methods by effectively alleviating data heterogeneity and accelerating model convergence through RACBA, while significantly reducing training costs via CDOA. Finally, the experimental results have demonstrated that C^2 -SFL has significantly outperformed the baseline methods in reducing latency and energy consumption.

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